

## P4 — Design-aware main estimate

Week 6 · applies L5 (Survey analysis) · optional and ungraded — no separate deadline, nothing to submit; M1 is assembled from these five increments Keep in your group GitHub repo: the reproduced main estimate + a design-aware re-estimate + a comparison.

### The question this week

*What is your best estimate of the main association once you respect how the data were actually generated?* First reproduce the paper’s headline number; then re-estimate it the way the study **design** demands, and explain any difference.

### What to do

1. **Define the estimand** in one sentence (which contrast, which outcome, what time horizon).
2. **Replicate the paper’s main model** exactly as they fit it.
3. **Respect the feature you named at M0 criterion 5.** For a complex survey, use the weights, strata and clusters — and **subset the design object**, never treat the analytic sample as a simple random sample; that gives you a design-aware re-estimate. For a time-to-event outcome, keep follow-up time and censoring in a survival model, don’t collapse to “ever died” — then interrogate the *specific* structural choice you named at M0 (time scale, censoring rule, index date, competing events) by refitting under a defensible alternative to it.
4. **Compare and interpret — but only what is comparable.** For a survey paper: point estimate, uncertainty and interpretation, naïve/published vs design-aware. That is a like-for-like contrast because the two fits are the same estimand on the same scale and differ only in the design. For a time-to-event paper with no weights there is no such pair: report the reproduced survival estimand, then what the alternative structural choice does to it, and say in one sentence what the paper’s choice assumed. Never set a survival model beside a binary “ever died” model and call the gap a design correction — those are different estimands on different scales, and the gap measures the collapse, not the design.

### Deliverable

- The replicated main estimate (matches the paper) → your M1 **Table 2 / KM slide**.
- The design-aware re-estimate + a side-by-side comparison table.
- 2–3 sentences on what changed and why — this feeds your **M1** synthesis, and it is the kind of design choice you will be asked to evaluate when you review another group’s M1 at M2.

If your paper is not a complex survey, the feature you respect is the one you named at M0 criterion 5 — follow-up time and censoring — and the deliverable is a different shape. You have no weight to switch on and off, so there is no naïve-vs-correct pair to table. What you keep in the repo is: the paper’s survival estimand, reproduced with follow-up time and censoring intact; a refit under a defensible alternative to the one structural choice you named at M0; and 2–3 sentences on what that choice was doing to the paper’s conclusion. That is a structural interrogation rather than a numeric correction — a different deliverable, not a lesser one.

### Worked example — MASLD/NHANES demonstration

- **Replicate first — the paper is *unweighted*.** An ordinary Cox model reproduces the headline: Group IV all-cause HR  $\approx 15.1$  **unadjusted** →  $\sim 2.9$  **fully adjusted** (the reproduction matches to 2–3 significant figures). Reproducing the *unweighted* result is the replicate step.
- **The improvement is the design the paper does not apply.** The cohort is the NHANES **fasting subsample** (FLI needs fasting triglycerides), so a design-aware estimate needs the **fasting-subsample weights (WTSAF)** pooled across six cycles, even though FLI also draws on non-fasting (GGT) labs.
- **Which weight is a rule, not a judgment call.** CDC’s least-common-denominator rule: when an analysis draws on variables from different components, use the weight of the **smallest** applicable

subsample. If your exposure needs a fasting analyte, the fasting weight governs — a larger-sample weight would weight to a population in which your exposure cannot be constructed. The exercise is to *name the straddle and apply the rule*, not to argue the choice open. Then report a WTMEC-weighted run beside it, **labelled as a sensitivity analysis** — it is not a defensible primary, but it shows how much the rule changed the answer.

- **Ask the comparison the paper didn't.** Estimate Group IV vs **II** and IV vs **III** directly, not only each vs Group I.

## What good looks like

- **Exceptional, survey route:** replicates the paper's model, then a correct design-aware re-estimate; names the weight/strata/cluster used and *why*; interprets the change.
- **Exceptional, time-to-event route:** reproduces the paper's survival estimand with follow-up time and censoring intact; names the structural choice identified at M0 and *why* it is contestable; refits under a defensible alternative and interprets what the paper's choice was doing. Offering a binary “ever died” model as the corrected estimate is not this band — it is the Weak band below.
- **Adequate:** main estimate reproduced; and either a reasonable design adjustment attempted (survey route), or the M0 structural choice named and probed even if the refit is incomplete (time-to-event route).
- **Weak:** refits with `svyglm` on “ever died,” discarding follow-up time and censoring; or ignores the design entirely.

## Be ready to defend (M3)

*“Which design feature did you respect, and why that one — and what turns on it?”*

For a survey paper the answer is a number: your headline estimate with the weight versus without it. For a time-to-event paper it is not a number: it is what the paper's structural choice assumed — the time scale, the censoring rule, the index date, how competing events were treated — and what a defensible alternative does to the conclusion. A binary “ever died” model is not an answer to this question.

## AI co-pilot note

A co-pilot often reaches for the *simplest* model that “works” — e.g. logistic regression on a binary “died” flag — because it avoids the survival/design machinery. That silently throws away follow-up time, censoring, and the sampling design. Insist on the model the data structure requires.

## Lab → project handoff

You are applying **L5 (Survey analysis)** — contrasting ordinary sample summaries with survey-design estimates, and subsetting the *design* rather than the data (EpiMethods `surveydataE`).